



Discrete Structures I

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Sets and Relations

- Sets and relations are the most fundamental parts of mathematics. Informally, sets are simply collections of objects. From sets we build every other discrete structure, including, but not limited to, combinations, functions, relations, and graphs.
- This topic will explore much of what sets has to offer. Sets as collections, sets as objects themselves, and relations as sets of pairs.
- Much as how propositional and predicate logic has set the framework for the semantics of mathematical language, sets set the framework for mathematical objects.
- Both logic and sets are the foundation on which the rest of discrete mathematics is built.



Introduction

- Most of mathematics is built up from the very basic concept of a set. It is so basic in fact, that the concept of a set is rarely defined formally rather something thought of intuitively as just “a collection of objects”.
- Defining sets formally is outside the scope of this course and is described by “axiomatic set theory” and, in particular, Zermelo-Fraenkel Set Theory.
- Rather, we will follow naive set theory, and use the previously mentioned intuitive definition.
- You must be familiar with the following sets
 - ① The set of natural numbers $\mathbb{N}$ are the “whole numbers”: $0, 1, 2, 3, \dots$
 - ② The set of integers $\mathbb{Z}$ are the whole numbers and their negatives: $\dots, -2, -1, 0, 1, 2, \dots$
 - ③ The set of real numbers $\mathbb{R}$ are “all” the numbers, including fractions, decimals, and irrational numbers like π
 - ④ The set of complex numbers $\mathbb{C}$ are all numbers of the form $a + bi$ where $i = \sqrt{-1}$ and a, b are real numbers.



Set Construction

- Constructing a set is the first and foremost part of set theory.
- Naturally, we cannot use sets, or perform operations on sets, without having a set in the first place. We will see 3 ways to construct a set:
 - 1 Roster Method
 - 2 Set Builder notation
 - 3 Interval Notation



Roster Method

The roster method for set construction is the most simple. We simply write down all of the members of a set between curly braces.

$A = \{a, b, c, d\}$ The set A contains four members a, b, c , and d

When there is an obvious pattern to the elements of a set, one can abbreviate the roster method using an ellipse (i.e. ...). For example, the set of positive integers less than 100 may be written as

$$\{1, 2, 3, \dots, 99\}$$

The set of all positive multiples of 3 could be written as:

$$\{3, 6, 9, 12, \dots\}$$

Notice that the set does not need to be finite to use the roster method.

Generally, sets can have a finite or infinite number of members, as we will see.



Roster Method

- Moreover, sets do not necessarily need to contain numbers or only numbers. Sets are collections of any “thing” or “object”.
- Those objects could be any combination of letters, strings, colors, shapes, matrices, etc. Sets can even contain other sets!. As an example, the following are also valid sets:

$$\{\text{"Alice"}, \text{"Ben"}, \text{"Carol"}, \text{"Dominic"}\}$$
$$\{1, 2, \text{"red"}, \text{"blue"}, \text{"fish"}, 4, 6\}$$
$$\{\{a, b\}, \text{"foo"}, 12, \{\text{"bar"}, 6\}\}$$
$$\{a, b, c, \dots, z\}$$



Roster method for sets of numbers

Using the roster method, we can describe some well-known mathematical sets as in the following example.

Example 1.1

Natural numbers: $\mathbb{N} = \{0, 1, 2, 3, \dots\}$

Integers: $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

Positive Integers: $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$



Set Builder

The second, and more general way of describing a set is using **set builder notation**. You have likely seen this notation used implicitly in previous math courses.

Definition 1.1: Set builder notation

Set builder notation specifies elements that should be included in a set based on a variable (respectively, some expression) and a condition on that variable (respectively, expression). The variable and condition are separated by a vertical bar |

$$\{x \mid \text{some condition on } x\}$$

$$\{\text{some expression} \mid \text{some conditions or properties of that expression}\}$$

In this notation the vertical bar | is read as “such that”. The following notation defines the set of positive integers less than 100 using set builder notation. It can be read as “the set containing x , where x is an integer, such that x is greater than 0 and less than 100”.

$$\{x \in \mathbb{Z} \mid 1 < x < 100\}$$



Set Builder

- This notation may have included a new symbol: $\in$
- This symbol means “in” or “belongs to” and denotes that an object is a member of a set.
- $x \in \mathbb{Z}$ can be read as “ x belongs to the integers” or “ x in the set of integers” or “ x is an integer”.
- The set of rational numbers or complex numbers are easily defined using set builder notation, but are quite complicated to define using the roster method.

Definition 1.2: Set of rational numbers and complex numbers

$$\mathbb{Q} = \left\{ \frac{m}{n} \mid m, n \in \mathbb{Z} \text{ and } n \neq 0 \right\}$$

$$\mathbb{C} = \left\{ a + bi \mid a, b \in \mathbb{R}, i^2 = -1 \right\}$$



Set Builder

- In set builder notation, conditions which should simultaneously be satisfied are often listed after the $|$ and separated by commas.
- Their conjunction is implied. Alternatively, one can explicitly write “and”.

Definition 1.3: Set builder and conjunctions

The following sets are equivalent.

$$\{x \in \mathbb{Z}^+ \mid x < 10 \wedge x \text{ is even} \}$$

$$\{x \in \mathbb{Z}^+ \mid x < 10 \text{ and } x \text{ is even} \}$$

$$\{x \in \mathbb{Z}^+ \mid x < 10, x \text{ is even} \}$$

$$\{2, 4, 6, 8\}$$



Natural language description

- Just as we can translate between Natural language and predicates, we can similarly use natural language in place of the explicit set builder notation.
- This is often called a semantic description of the set, where a sentence is used to describe the properties of objects contained in a set.
- Set builder notation has one strong benefit over the roster method: you do not need to know the precise members of a set in order to construct the set.

Definition 1.4: Semantic descriptions

The following are valid semantic descriptions of sets.

Let A be the set of three primary colors.	$A = \{\text{red, blue, yellow}\}$
Let B be the set of five smallest positive integers.	$B = \{1, 2, 3, 4, 5\}$
Let C be the set of positive rational numbers with 1 as a numerator.	$C = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$



Interval Notation

- For sets of real numbers, a particularly useful notation is interval notation
- For two real numbers a and b such that $a < b$ the following are possible intervals

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$$

$$(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$$

$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$$

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

- The choice of including the end point or an interval or not determines if the interval is closed or open.



Interval Notation

Definition 1.5: closed interval

A closed interval is the set of all numbers between two end points, including those endpoints. It is denoted by square brackets. $[a, b]$ is all numbers x such that $a \leq x \leq b$

Definition 1.6: open interval

An open interval is the set of all numbers between two end points, excluding those endpoints. It is denoted by round brackets. (a, b) is all numbers x such that $a < x < b$



Set Properties

Now that we know how to construct sets, it is time to discuss particular sets and their properties.



Membership and Equality

- Membership is the basic property of sets from which all other operations and properties can be defined.
- A set either contains an object or does not contain an object. Sets define a clear and unambiguous description of its members.
- The notation $\in$ used earlier say that an element is included in the set and the notation $\notin$ says an element is not included in a set.

$$a \in \{a, b, c, d, e\}$$

$$4 \notin \{1, 3, 5, 7, 9, \dots\}$$



- Membership is strict binary. An object is either in a set or is not in a set. Because of that, we have the following interesting results.

$$1 \in \{1, 2\}$$

$$1 \in \{2, 1\}$$

$$1 \in \{1, 1, 1, 2\}$$

$$1 \in \{1, 1, 1, 2, 2, 27\}$$

$$3 \notin \{1, 1, 1, 2\}$$



Equality of Sets

Definition 1.7: Equality of Sets

Two sets A and B are **equal** if every member of A is a member of B and every element of B is a member of A

- Stated in predicate logic, two sets are equal if $\forall x(x \in A \leftrightarrow x \in B)$
- Because the only basic property of a set is membership, a set does not care about the order in which elements are listed or the number of times each element appears in the set construction. For example, the following sets are equal.

Example 1.2: Set Equality

$$\{1, 2\} = \{2, 1\} = \{1, 1, 2\} = \{2, 1, 2, 1, 1, 1, 1, 2, 1, 2, 1, 2, 2\}$$

To avoid this awkward and “non-canonical” description of sets, it is often said that sets should only contain unique, distinct, or distinguishable objects

In particular, when using the roster method, one should only write each unique element once. Therefore $\{1, 2\}$ or $\{2, 1\}$ is the most “correct” description of the previous set.



Sets in Python

- In Python, sets are a particular kind of data structure.
- They follow much of the semantics of the mathematical set. Sets are generic collections of objects.
- However, for many reasons associated with simplicity and efficiency of programming, Python requires a canonical form for their sets.

```
set1 = {'a', 'b', 'c'}  
set2 = {'b', 'c', 'c', 'a', 'c', 'b'}  
print("set1:", set1)  
print("set2:", set2)  
print(set1 == set2);
```

#Output

```
set1: {'a', 'b', 'c'}  
set2: {'a', 'b', 'c'}  
True
```



Special Sets: Empty and Universal

Definition 1.8: Empty Set $\emptyset$

The empty set is the unique set containing no elements. It is denoted by $\emptyset$

- In Roster method the $\emptyset$ is denoted by $\{\}$
- In contrast with the empty set, the universal set is the set which contains every element.
- For practical purposes, we can instead consider a particular universe of things. That is, a universe or domain of discourse. Indeed, we have already seen the set U to represent the domain of discourse in predicate logic.
- Such a universe can be implicit, for example the set containing days of the week

$$D = \{\text{Monday, Wednesday, Friday}\}$$

$$U = \{\text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}\}$$



True or False

Are the following equalities true or false?

① $\emptyset = \{\}$

② $\emptyset = \{x \in \mathbb{Z} \mid 2x = 3\}$

③ $\emptyset = \{\{\}\}$

④ $\emptyset = \{x \in \mathbb{R} \mid x^2 + 2 = 0\}$

⑤ $\emptyset = \{\emptyset\}$



Subsets

Definition 1.9

A set A is a subset of the set B , written $A \subseteq B$, if every member of A is also a member of B .

This definition is a one-way relation. Unlike equality, a subset only requires that every member of the subset is also a member of the **superset**.

Example 1.3: Subsets

The following are valid subsets

$$\{1, 2, 3\} \subseteq \{1, 2, 3, 4, 5\}$$

$$\{c, e\} \subseteq \{a, b, c, e, y, z\}$$

$$\{2, 4, 6\} \subseteq \{2, 4, 6\}$$

$$\{\text{foo}, \text{bar}\} \subseteq \{\text{bar}, \text{buzz}, \text{foo}\}$$



Equality of Sets

By the definition of subset, we have $\emptyset \subseteq A$ for any set A . Indeed, since $\emptyset$ has no element, it is the subset of anything.

Notice also that by definition the symbol $\subseteq$ is similar to $\leq$ and means "is a subset or is equal to". Indeed two sets A and B are equal if $A \subseteq B \wedge B \subseteq A$

Example 1.4: Equality of Sets

Let $A = \{1, 3, 5, 7, 9\}$. Let $B = \{3, 5\}$. Let C be the positive odd integers less than 10. Let D be the positive odd integers less than 100

We have: $B \subseteq A$, $B \subseteq C$, $B \subseteq D$

We also have: $A \subseteq C$, $A \subseteq D$, $C \subseteq A$, $C \subseteq D$

Therefore: $A = C$



Size of a Set

- The last basic property of a set is its size.
- Sets are crucial in counting and combinatorics, so being able to discuss their size concretely is important.
- The terms “cardinality” and “size” are often used interchangeably.

Definition 1.10: Cardinality

The cardinality of a finite set is the number of distinct elements contained in the set. For a set A its cardinality is denoted $|A|$

- The cardinality of any set is non-negative. When the cardinality of a set is some non-negative integer n we say that the set is *finite* otherwise infinite



Size of a Set

Definition 1.11: Singleton

A singleton is a set with cardinality one. That is, it is a set with exactly one member.



Cardinality Exercise

What is the cardinality of the following sets?

$$A = \{1, 2, 3, 4, 5\} \quad (1)$$

$$B \text{ is the set of all letters in the English alphabet} \quad (2)$$

$$C = \{x \in \mathbb{Z} \mid x < 100 \text{ and } 3x > -33\} \quad (3)$$



Power Set

Definition 1.12: Power Set

The power set of a set A is the set of all possible subsets of A .
The power set of A is denoted by $\mathcal{P}(A)$.

In Set builder Notation, $\mathcal{P}(A) = \{S \mid S \subseteq A\}$

Example 1.5: A Power Set

Let $A = \{a, b, c\}$. The members of $\mathcal{P}(A)$ are $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

Theorem 1.1

For a finite set A with $|A| = n$, $|\mathcal{P}(A)| = 2^n$



Operations on Sets

- There are various ways to manipulate, combine, and operate on sets to produce another. Some operations even produce multiple sets.
- Before we embark on the set operations in particular look graphical representation of sets: Venn diagrams.
- In a Venn diagrams, a set is represented by a circle. It may or may not have its members indicated within the circle by labelled dots. If we want to be explicit about the universe from which a set is constructed, we draw the Venn diagram inside of a large rectangle. Circles are sets, dots are members, and a rectangle is the universe.



Figure 1: The set $A = \{1, 2, 3, 4, 5\}$ in a universe U represented as a Venn diagram.



Subset

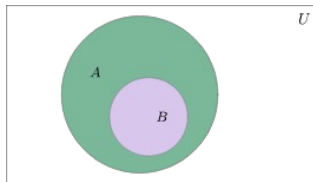


Figure 2: The set A and B with $B \subseteq A$ in a universe U represented as a Venn diagram.



Union and Intersection

Definition 1.13: Set Union

The union of two sets A and B is the set of elements which are members of A or members of B , or members of both. The union of A and B is denoted $A \cup B$ as shown in figure 3

In setbuilder notation, the union of two sets A and B can be written as $\{x \mid x \in A \vee x \in B\}$

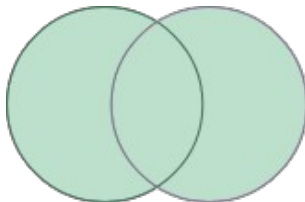


Figure 3: The union of sets A and B is everything in A or B or both



Set Intersection

Definition 1.14: Set Intersection

The intersection of two sets A and B is the set of elements which are members of both A and B . The intersection of A and B is denoted $A \cap B$ as shown in the figure 4

In set builder notation, the intersection of two sets A and B can be written as: $\{x \mid x \in A \wedge x \in B\}$

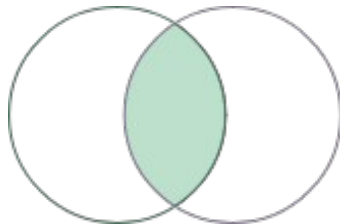


Figure 4: The union of sets A and B is members in both A and B



Union and Intersection

Example 1.6: Union and Intersection

Let $A = \{1, 2, a, b, c\}$ and $B = \{2, 4, 6, x, a\}$

$$A \cup B = \{1, 2, 4, 5, a, b, c, x\}$$

$$A \cap C = \{2, a\}$$

- Union and intersection are both associative and commutative operations. Therefore, given multiple sets, say A_1, A_2, A_3 , the following are equivalent.

$$(A_1 \cup A_2) \cup A_3 = A_1 \cup (A_2 \cup A_3) = A_1 \cup A_2 \cup A_3$$

$$(A_1 \cap A_2) \cap A_3 = A_1 \cap (A_2 \cap A_3) = A_1 \cap A_2 \cap A_3$$



Union and Intersection

Where there are many sets, use $\bigcup$ and $\bigcap$ to represent union or intersection over a sequence of sets. This is akin to $\sum$ for summation and $\prod$ for product.

$$A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_n = \bigcup_{i=1}^n A_i$$

$$A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n = \bigcap_{i=1}^n A_i$$

Example 1.7: Union and Intersection, together

Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7, 8, 9\}$, and $C = \{1, 3, 5, 7, 9\}$

$$(A \cup B) \cap C = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} \cap C = \{1, 3, 5, 7, 9\}$$

$$A \cup (B \cap C) = A \cup \{5, 7, 9\} = \{1, 2, 3, 4, 5, 7, 9\}$$

NOTE: Although union and intersect are individually associative, they do not mix. Indeed, $(A \cup B) \cap C$ in general, does not equal $A \cup (B \cap C)$



Disjoint Sets

Definition 1.15: Disjoint Sets

Sets A and B are said to be disjoint if $A \cap B = \emptyset$



Set union and intersect

Let $A = \{a, b, y, a, c\}$, $B = \{4, 5, 6, c, z\}$, $C = \{a, 2, 3, 4, 5, b\}$,
 $D = \{1, 2, 3, 4\}$

What is the result of the following unions and intersections?

- ① $A \cup B$
- ② $A \cup B \cup C$
- ③ $A \cup C$
- ④ $A \cap D$
- ⑤ $(A \cup C \cup B) \cap D$

Solution



Set union and intersect

Let $A = \{a, b, y, a, c\}$, $B = \{4, 5, 6, c, z\}$, $C = \{a, 2, 3, 4, 5, b\}$,
 $D = \{1, 2, 3, 4\}$

What is the result of the following unions and intersections?

- ① $A \cup B$
- ② $A \cup B \cup C$
- ③ $A \cup C$
- ④ $A \cap D$
- ⑤ $(A \cup C \cup B) \cap D$

Solution

- ① $\{a, b, c, y, z, 4, 5, 6\}$



Set union and intersect

Let $A = \{a, b, y, a, c\}$, $B = \{4, 5, 6, c, z\}$, $C = \{a, 2, 3, 4, 5, b\}$,
 $D = \{1, 2, 3, 4\}$

What is the result of the following unions and intersections?

- ① $A \cup B$
- ② $A \cup B \cup C$
- ③ $A \cup C$
- ④ $A \cap D$
- ⑤ $(A \cup C \cup B) \cap D$

Solution

- ① $\{a, b, c, y, z, 4, 5, 6\}$
- ② $\{a, b, c, y, z, 2, 3, 4, 5, 6\}$



Set union and intersect

Let $A = \{a, b, y, a, c\}$, $B = \{4, 5, 6, c, z\}$, $C = \{a, 2, 3, 4, 5, b\}$,
 $D = \{1, 2, 3, 4\}$

What is the result of the following unions and intersections?

- 1 $A \cup B$
- 2 $A \cup B \cup C$
- 3 $A \cup C$
- 4 $A \cap D$
- 5 $(A \cup C \cup B) \cap D$

Solution

- 1 $\{a, b, c, y, z, 4, 5, 6\}$
- 2 $\{a, b, c, y, z, 2, 3, 4, 5, 6\}$
- 3 $\{a, b, c, y, 2, 3, 4, 5\}$



Set union and intersect

Let $A = \{a, b, y, a, c\}$, $B = \{4, 5, 6, c, z\}$, $C = \{a, 2, 3, 4, 5, b\}$,
 $D = \{1, 2, 3, 4\}$

What is the result of the following unions and intersections?

- ① $A \cup B$
- ② $A \cup B \cup C$
- ③ $A \cup C$
- ④ $A \cap D$
- ⑤ $(A \cup C \cup B) \cap D$

Solution

- ① $\{a, b, c, y, z, 4, 5, 6\}$
- ② $\{a, b, c, y, z, 2, 3, 4, 5, 6\}$
- ③ $\{a, b, c, y, 2, 3, 4, 5\}$
- ④ $\emptyset$



Set union and intersect

Let $A = \{a, b, y, a, c\}$, $B = \{4, 5, 6, c, z\}$, $C = \{a, 2, 3, 4, 5, b\}$,
 $D = \{1, 2, 3, 4\}$

What is the result of the following unions and intersections?

- ① $A \cup B$
- ② $A \cup B \cup C$
- ③ $A \cup C$
- ④ $A \cap D$
- ⑤ $(A \cup C \cup B) \cap D$

Solution

- ① $\{a, b, c, y, z, 4, 5, 6\}$
- ② $\{a, b, c, y, z, 2, 3, 4, 5, 6\}$
- ③ $\{a, b, c, y, 2, 3, 4, 5\}$
- ④ $\emptyset$
- ⑤ $\{2, 3, 4\}$



Set union and intersect

Let $A = \{a, b, y, a, c\}$, $B = \{4, 5, 6, c, z\}$, $C = \{a, 2, 3, 4, 5, b\}$,
 $D = \{1, 2, 3, 4\}$

What is the result of the following unions and intersections?

- ① $A \cup B$
- ② $A \cup B \cup C$
- ③ $A \cup C$
- ④ $A \cap D$
- ⑤ $(A \cup C \cup B) \cap D$

Solution

- ① $\{a, b, c, y, z, 4, 5, 6\}$
- ② $\{a, b, c, y, z, 2, 3, 4, 5, 6\}$
- ③ $\{a, b, c, y, 2, 3, 4, 5\}$
- ④ $\emptyset$
- ⑤ $\{2, 3, 4\}$



Important Set Identities

Set union and intersection also have some important identities.

- ① $A \cup \emptyset = A$
- ② $A \cap \emptyset = \emptyset$
- ③ $A \cup A = A \cap A = A$
- ④ $(A \cup B) \cap A = A$
- ⑤ $(A \cap B) \cup A = A$

The latter two are the absorption laws for sets. Earlier seen in propositional logic. Visually, absorption for sets (fig. 5) is easier to understand than in propositional logic.

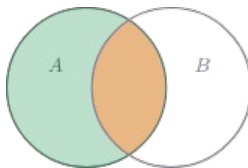


Figure 5: The intersection of sets $A \cap B$ is shaded orange. Its Union with A is A itself



Inclusion-Exclusion Principle

Proposition 1.1: Cardinality of $A \cup B$

For two sets A and B the cardinality of $A \cup B$ is given by

$$|A \cup B| = |A| + |B| - |A \cap B|$$

The inclusion-exclusion principle states that the size of two sets together is the size of each set individually, minus the size of their overlap - to avoid double counting.



Set Difference and Complement

Definition 1.16: Set Difference

The difference between two sets A and B (fig 6) is the set of elements which are members of A which are not members of B . The difference of A by B is denoted $A \setminus B$.

In set builder notation, the set difference $A \setminus B$ can be written as:
 $\{x \mid x \in A \wedge x \notin B\}$

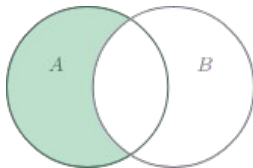


Figure 6: The Set difference $A \setminus B$ is everything in A that is not in B



Set Difference and Complement

Definition 1.17: Set Complement

The complement of a set A is the set of elements which are not members of A . The set complement is denoted by $\overline{A}$ or A^c .

In set builder notation, the set complement $\overline{A}$ (fig: 7) can be written as:
 $\{x \in U \mid x \notin A\}$

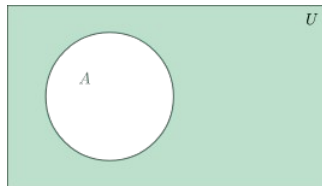


Figure 7: The complement of a set A is everything not in A



Set Difference and Complement

Example 1.8

Let $A = \{a, b, c\}$, $B = \{1, 2, 3\}$, $C = \{x \in \mathbb{Z} \mid |x - 2| \geq 4\}$

$$A \setminus \{a, b\} = \{c\}$$

$$A \setminus B = \{a, b, c\} = A$$

$$B \setminus \mathbb{Z} = \emptyset$$

$$\overline{C} = \{-1, 0, 1, 2, 3, 4, 5\}$$

Notice that the universe for C is the set of integers.



Set Difference and Complement Identities

Proposition 1.2: Set Difference and Complement Identities

- ① $A \setminus B = A \cap \overline{B}$
- ② $(A^c)^c = A$
- ③ $\overline{\emptyset} = U$
- ④ $A \setminus \emptyset = A$
- ⑤ $A \setminus U = \emptyset$

What is the result of the following difference and complement?

- ① $\{x \in \mathbb{Z}^+ \mid x \text{ is even}\} \setminus \{4, 8, 12, 16, \dots\}$
- ② $\{x \in \mathbb{Q} \mid x \notin \mathbb{Z}\}^c$

Solution



Set Difference and Complement Identities

Proposition 1.3: Set Difference and Complement Identities

- ① $A \setminus B = A \cap \overline{B}$
- ② $(A^c)^c = A$
- ③ $\overline{\emptyset} = U$
- ④ $A \setminus \emptyset = A$
- ⑤ $A \setminus U = \emptyset$

What is the result of the following difference and complement?

- ① $\{x \in \mathbb{Z}^+ \mid x \text{ is even}\} \setminus \{4, 8, 12, 16, \dots\}$
- ② $\{x \in \mathbb{Q} \mid x \notin \mathbb{Z}\}^c$

Solution

- ① $\{2, 6, 10, 14, 18, \dots\}$
- ② $\mathbb{Z}$. The original set is all rational numbers which are not integers (where $U = \mathbb{Q}$). The complement of that is surely all integers.



Symmetric Difference

Definition 1.18: Symmetric Difference

The symmetric difference of two sets A and B is the set of elements which are members of A or members of B but not both. The symmetric difference of A and B is denoted by $A \oplus B$

- The “symmetric” in symmetric difference comes from the following identity:

$$A \oplus B = (A \setminus B) \cup (B \setminus A)$$

- A more intuitive way to think about symmetric difference is as the union of two sets minus their intersection.

$$A \oplus B = (A \cup B) \setminus (A \cap B)$$

- The venn diagram for the above relation is fig 8



Symmetric Difference

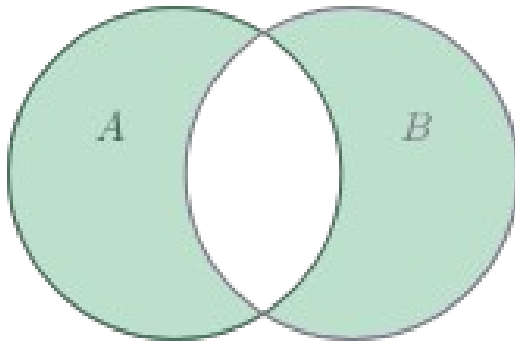


Figure 8: The symmetric difference of two sets A and B is their union minus their intersection



Cartesian Product

The final set operation to be considered is their Cartesian product. But first, we need a new kind of discrete structure.

Definition 1.19: Tuple

A tuple is a finite ordered sequence of elements (from a set). An n -tuple is a tuple with n elements.

Special names for tuples of certain lengths:

- 1 a 1-tuple is a monuple or a singleton,
- 2 a 2-tuple is a couple or a pair,
- 3 a 3-tuple is a triple or triplet,
- 4 a 4-tuple is a quadruple,
- 5 a a -tuple is a pentuple, and so on ...

Tuples are denoted using parentheses. For example, (a, b) is a tuple, (c, b, a) and (c, c, a, b, a, c) is a tuple. Unlike sets, ordering is important for tuples.



Cartesian Product

- The elements of a tuple are called coordinates. in a tuple (a, b, c) , a is the first coordinate, b is the second coordinate, and c is the third coordinate.
- Two tuples are equal if and only if every coordinate is equal.

$$(a_1, b_1) = (a_2, b_2) \leftrightarrow a_1 = a_2 \wedge b_1 = b_2$$

Definition 1.20: Tuple Equality

The tuples $(3, 7, 5)$ and $(3, 7, 5)$ are equal. However, $(3, 5, 7) \neq (3, 7, 5)$.

- Tuples will become increasingly important in Relations and Functions.



Cartesian Product

Definition 1.21: Cartesian Product

The Cartesian product of two sets A and B is the set of all pairs (a, b) where $a \in A$ and $b \in B$. The Cartesian product is denoted $A \times B$.

- The Cartesian product $A \times B$ is read “ A cross B ”.
- The Cartesian product is also called a direct product.
- The result of a Cartesian product is all possible pairs where the first coordinate ranges over all elements of A and the second coordinate ranges over all elements of B .
- We can define the Cartesian product using set builder notation:

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

- Notice that if either A or B is the empty set, then $A \times B = B \times A = \emptyset$. For two non-empty sets, unless $A = B$, $A \times B \neq B \times A$



Cartesian Product

Example 1.9: Cartesian Product

Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$

$$A \times B = \{(1, x), (2, x), (3, x), (1, y), (2, y), (3, y)\}$$

$$B \times A = \{(x, 1), (y, 1), (x, 2), (y, 2), (x, 3), (y, 3)\}$$

One way of viewing the Cartesian product $A \times B$ is as a table. The columns range over all possible values of A and the rows range over all possible values of B . If $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$ then fig: 9

	x	y	z
1	$(1, x)$	$(1, y)$	$(1, z)$
2	$(2, x)$	$(2, y)$	$(2, z)$
3	$(3, x)$	$(3, y)$	$(3, z)$

Figure 9: Cartesian Product of $A = \{1, 2, 3\}$ Cross $B = \{x, y, z\}$



Proofs with Sets

- Sets, being fundamental to every area of mathematics, many proofs rely on arguments related to set membership.
- This section examines how to **use sets and set membership** to prove some theorems.
- The first step to most proofs involving sets is to remember the definition of set equality.
- Two sets A and B are equal when we have $x \in A \leftrightarrow x \in B$.
- To prove this biconditional, we must show $x \in A \rightarrow x \in B$ and $x \in B \rightarrow x \in A$.

Proposition 1.4

If $A \subseteq B$ then $A \cup B = B$



Proofs with Sets

Proof of Proposition 1.4.

We look to prove $A \subseteq B \rightarrow A \cup B = B$

Assume $A \subseteq B$. We look to prove $x \in A \cup B \leftrightarrow x \in B$

- ① $(\rightarrow)$ Assume $x \in A \cup B$. Then $x \in A$ or $x \in B$. Proceed by cases:
 - If $x \in B$ we are done
 - If $x \in A$, we know $A \subseteq B$, therefore $x \in B$
- ② $(\leftarrow)$ Assume $x \in B$. Therefore x must surely be in $A \cup B$.





Proofs with Sets

Proposition 1.5: Proving Set Distribution

For any sets A, B, C , $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof.

Let $X = A \cup (B \cap C)$ and $Y = (A \cup B) \cap (A \cup C)$. Since we are proving a set equality, we must prove a biconditional: $\forall x (x \in X \leftrightarrow x \in Y)$

① $(\rightarrow)$ First we prove that $x \in X \rightarrow x \in Y$

Assume $x \in X$. By union, we have that $x \in A$ or $x \in (B \cap C)$. Proceed by cases:

- If $x \in A$ then surely $x \in A \cup B$ and $x \in A \cup C$. Therefore $x \in Y$
- If $x \in (B \cap C)$ then $x \in B$ and $x \in C$. Thus, x belongs to both $A \cup B$ and $A \cup C$. Therefore $x \in Y$



Proofs with Sets

Proof Cont'd.

② ($\leftarrow$) We now prove $x \in Y \rightarrow x \in X$

Assume $x \in Y$. Therefore x is a member of both $A \cup B$ and $A \cup C$

By union of $A \cup B$

- If $x \in A$ the surely $x \in A \cup (B \cap C)$ and $x \in X$
- If $x \notin A$ then $x \in B$. We also have $x \in A \cup C$; but since $x \notin A$ then $x \in C$

Hence, $x \in B \cap C$ and surely $x \in X$





Exercises

Write out the following sets using the roster method.

Let $Q(x, y) := x(2y + 4) = 0$

- ① $\{x \mid x \in \mathbb{Z}^+ \text{ and } x < 5\}$
- ② $\{x \in \mathbb{Q} \mid 2x \in \mathbb{Z}\}$
- ③ $\{x \mid x, y \in \mathbb{Z}, \forall y Q(x, y)\}$
- ④ $\{y \mid x, y \in \mathbb{Z}, \exists x Q(x, y)\}$

Solution



Exercises

Write out the following sets using the roster method.

Let $Q(x, y) := x(2y + 4) = 0$

- ① $\{x \mid x \in \mathbb{Z}^+ \text{ and } x < 5\}$
- ② $\{x \in \mathbb{Q} \mid 2x \in \mathbb{Z}\}$
- ③ $\{x \mid x, y \in \mathbb{Z}, \forall y Q(x, y)\}$
- ④ $\{y \mid x, y \in \mathbb{Z}, \exists x Q(x, y)\}$

Solution

- ① $\{1, 2, 3, 4\}$
- ② $\{\dots, \frac{-3}{2}, -1, \frac{-1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$
- ③ $\{0\}$
- ④ $\{\dots, -2, -1, 0, 1, 2, \dots\}$



Exercises

Let $A = \{x \mid x \text{ is an odd integer and } 2x + 5 > 20\}$

Let $B = \{x \in \mathbb{Z}^+ \mid x > 7 \text{ and } \exists k \in \mathbb{Z} x = 2k + 1\}$

Is A a subset of B ? Is B a subset of A

Solution



Exercises

Let $A = \{x \mid x \text{ is an odd integer and } 2x + 5 > 20\}$

Let $B = \{x \in \mathbb{Z}^+ \mid x > 7 \text{ and } \exists k \in \mathbb{Z} x = 2k + 1\}$

Is A a subset of B ? Is B a subset of A

Solution

Both. $A = B$ Indeed, we have

$A = \{9, 11, 13, 15, \dots\}$ and

$B = \{9, 11, 13, 15, \dots\}$



Exercises

Let $U = \{x \in \mathbb{Z}_{\geq 0} \mid x \leq 10\}$. Let $A = \{1, 2, 3, 4, 5\}$. Let $B = \{4, 5, 6, 7, 8\}$

Write out the following sets using the roster method.

① $A \cup B$

② $A \cap B$

③ $\overline{A}$

④ $\overline{B}$

⑤ $A \setminus B$

⑥ $B \setminus A$

Solution



Exercises

Let $U = \{x \in \mathbb{Z}_{\geq 0} \mid x \leq 10\}$. Let $A = \{1, 2, 3, 4, 5\}$. Let $B = \{4, 5, 6, 7, 8\}$

Write out the following sets using the roster method.

① $A \cup B$

② $A \cap B$

③ $\overline{A}$

④ $\overline{B}$

⑤ $A \setminus B$

⑥ $B \setminus A$

Solution

① $\{1, 2, 3, 4, 5, 6, 7, 8\}$

② $\{4, 5\}$

③ $\{0, 6, 7, 8, 9, 10\}$

④ $\{0, 1, 2, 3, 9, 10\}$

⑤ $\{1, 2, 3\}$

⑥ $\{6, 7, 8\}$



Exercises

Prove the following proposition.

Proposition 1.6

If $A \subseteq B$ then $A \cap B = A$



Exercises

Prove the following proposition.

Proposition 1.7

If $A \subseteq B$ then $A \cap B = A$

We look to prove $A \subseteq B \rightarrow A \cap B = A$

Proof.

Assume $A \subseteq B$ we prove $x \in A \cap B \leftrightarrow x \in A$

- ① ($\rightarrow$). Assume $x \in A \cap B$. Then $x \in A$
- ② ($\leftarrow$). Assume $x \in A$. We look to prove $x \in A \cap B$, and thus $x \in A$ and $x \in B$
The former holds by assumption. For the latter, we have $x \in A$ and $A \subseteq B$,
Therefore, it must be that $x \in B$

□



Exercises

Prove the following Proposition

Proposition 1.8

Let A , B , and C be sets $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Proof.

Let $X = A \cap (B \cup C)$ and $Y = (A \cap B) \cup (A \cap C)$

Since we are proving a set equality, we must prove a biconditional $\forall x (x \in X \leftrightarrow x \in Y)$

- ① $(\rightarrow)$. We first prove that $x \in X \rightarrow x \in Y$. Assume $x \in X$. We must prove x is in $A \cap B$ or $A \cap C$.

By intersect, we have that $x \in A$ and $x \in (B \cup C)$. By union, proceed in cases:

- If $x \in B$ then $x \in A \cap B$, therefore $x \in Y$
- If $x \in C$ then $x \in A \cap C$, therefore $x \in Y$

- ② $(\leftarrow)$. We now prove $x \in Y \rightarrow x \in X$. Assume $x \in Y$. We must prove x is in A and x is in $B \cup C$.

By Union, x is a member of either $A \cap B$ or $A \cap C$. Proceed by cases:

- If $x \in A \cap B$ then $x \in A$ and $x \in B$. Since $x \in B$ we have $x \in B \cup C$ and therefore $x \in X$
- If $x \in A \cap C$ then $x \in A$ and $x \in B$. Since $x \in C$ we have $x \in B \cup C$ and therefore $x \in X$

□



Relations

- Relations, and in particular, binary relations, are a very general class of objects.
- They generalize the notion of functions and can be used to encode a variety of things.
- Relations may encode transformations, mappings, or functions.
- They may also be used more generally to encode properties about a set of objects.



Binary Relations

- Given two sets A and B , their Cartesian product $A \times B$ is the set of all pairs (a, b) with $a \in A$ and $b \in B$.
- A binary relation is a subset of the Cartesian product, where membership in the relation set denotes that a relates to b .

Definition 2.1

Let A and B be sets. A binary relation from A to B is a subset of $A \times B$. A binary relation on A is a subset $A \times A$

- Notice that by definition, the empty set $\emptyset$ is a relation. So is the entire Cartesian product $A \times B$.
- However, these are both not very interesting.
- Our interest is non-empty proper subsets of the Cartesian product. .



- Consider the set S of students in this class, and set M majors available in the faculty of science at JKUAT.
- Then a binary relation from S to M would “connect” or “relate” students in S and majors in M .
- A critical piece of understanding relations is that they do not require that every element relates to another.
- Moreover, they do not require that every element relates to exactly one other.
- Relations are not necessarily functions.

Example 2.1

Let S be the set of students

$\{\text{Andrea, Bruce, Davood, Fiona, Jalal, Rahul}\}$

Let M be the set of majors

$\{\text{Computer Science, Mathematics, Physics, Chemistry, Biology}\}$



Example: Cont'd

A relation $\mathcal{R}$ from S to M is a subset of $S \times M$. Let $\mathcal{R}$ be as follows.

$$\begin{aligned}\mathcal{R} = \{ & \\ & (\text{Davood}, \text{Physics}), \\ & (\text{Andrea}, \text{Computer Science}), \\ & (\text{Rahul}, \text{Chemistry}), \\ & (\text{Bruce}, \text{Biology}), \\ & (\text{Fiona}, \text{Computer Science}), \\ & (\text{Andrea}, \text{Mathematics}), \\ & (\text{Bruce}, \text{Chemistry}), \\ & (\text{Rahul}, \text{Physics}), \\ & (\text{Rahul}, \text{Computer Science}) \\ & \}\end{aligned}$$



Example: Cont'd

Membership in $\mathcal{R}$ can be understood to encode that student $s \in S$ is majoring in major $m \in M$.

Notice this relation is not “one-to-one” (something we will define formally later in Functions).

This relation tells us that Davood is taking Physics, Andrea is taking Computer Science and Mathematics, Bruce is taking Biology and Chemistry, etc.



Relations as tables

Example 2.2: Relations as tables

Let $A = \{1, 2, 3\}$ and $B = \{a, b, c, d, e\}$. A binary relation from A to B could be

$$\{(1, a), (1, c), (1, d), (2, a), (2, b), (3, c), (3, e)\},$$

and its representation as a table:

	a	b	c	d	e
1	x		x	x	
2	x	x			
3			x		x

Figure 10: “X marks the spot”



Relations as Predicates

- Relations are one possible encoding of predicates.
- In particular, of propositional functions with two or more variables.
- A pair being a member of a binary relation encodes that binding the variables of a propositional function to that pair results in true.
- Let U be the set of all students of JKUAT. That is U is the domain of discourse. Let $F(x, y)$ be the propositional function " x and y are friends". Let $\mathcal{R}$ be a binary relation of U if:

$$(x, y) \in \mathcal{R} \leftrightarrow F(x, y),$$

We can say $\mathcal{R}$ represents $F(x, y)$



Constructing relations

Example 2.3: Set builder for relations

Consider the set of all fractions (rational numbers) which, when reduced fully, are actually integers. Such a property is actually a relation between the numerator and the denominator. It says that, for a fraction $\frac{x}{y}$, x is a multiple of y

This relationship can be encoded as a binary relation $\mathcal{R}$ on $\mathbb{Z}$

$$\mathcal{R} = \{(x, y) \mid x, y \in \mathbb{Z}, \frac{x}{y} \in \mathbb{Z}\}, \quad \text{or}$$

$$\mathcal{R} = \{(x, y) \mid x, y \in \mathbb{Z}, \exists k \in \mathbb{Z} \ x = ky\}$$



Properties of Relations

There are many possible properties that relations may have. In this section we look at some important properties for binary relations on some set A .

That is the binary relation $\mathcal{R} \subseteq A \times A$

- ① Reflexivity
- ② Symmetry
- ③ Antisymmetry
- ④ Transitivity



1. Reflexivity

Definition 2.2: Reflexive

A binary relation $\mathcal{R}$ on a set A is reflexive if and only if, for every $a \in A$:

$$(a, a) \in \mathcal{R}.$$

Reflexivity means that every element must relate to itself. Equality is an obvious reflexive relation. For any number x , $x = x$

Example 2.4: Reflexive relations

Let P be the set of all people. Let $\mathcal{R}$ be the binary relation P relating all people with the same name.

It is clear that $\mathcal{R}$ is reflexive because someone name "Foo Bar" shares their first name with themselves, namely, "Foo"

Another way of thinking about a reflexive binary relation $\mathcal{R}$ on A is that, it is a superset of the set

$$\{(a, a) \mid a \in A\}$$

That is for any $a \in A$, the pair (a, a) is a member of $\mathcal{R}$



2. Symmetry

Definition 2.3: Symmetry

A Binary relation $\mathcal{R}$ on a set A is symmetric if and only if, for every $a, b \in A$

$$(a, b) \in \mathcal{R} \rightarrow (b, a) \in \mathcal{R}.$$

In this case, the ordering of the pair elements does not matter. Recall with a tuple, the ordering typically matters

Example 2.5

The following are symmetric relations.

$$\mathcal{R}_1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 5\} \mathcal{R}_2 = \{(a, b) \in \mathbb{Z}^2 \mid 2x + 2y = 4\}$$

Let U be all students at JKUAT and $P(x, y) := "x \text{ and } y \text{ are in the same class}$

$$\mathcal{R}_3 = \{(x, y) \mid P(x, y)\}$$



3. Antisymmetry

Definition 2.4: Antisymmetry

A binary relation $\mathcal{R}$ on a set A is antisymmetric if and only if for every $a, b \in A$

$$(a, b) \in \mathcal{R} \text{ and } (b, a) \in \mathcal{R} \rightarrow a = b.$$

This implies that any elements that relate symmetrically are actually equal.

Example 2.6: Antisymmetric Relation

Let $\mathcal{R} = \{(x, y) \in \mathbb{R}^2 \mid x \leq y\}$

Let a, b be any two real numbers. Notice that if $a \leq b$, then $(a, b) \in \mathcal{R}$.

If it is also the case that $b \leq a$ then $(b, a) \in \mathcal{R}$

However, notice that $(a \leq b) \wedge (b \leq a)$ can only occur when $a = b$.
Therefore $\mathcal{R}$ is antisymmetric



4. Transitivity

Definition 2.5: Transitivity

A binary relation $\mathcal{R}$ on a set A is transitive if and only if, for every $a, b, c \in A$

$$(a, b) \in \mathcal{R} \text{ and } (b, c) \in \mathcal{R} \rightarrow (a, c) \in \mathcal{R}.$$

Example 2.7: Transitive Relation

Let $\mathcal{R} = \{(a, d), (b, c), (b, d), (c, c), (c, d), (a, c)\}$.

This binary relation is transitive. The only pairs that fulfill the hypothesis of transitivity are (a, c) and (c, d) . Also (a, b) . What others can you identify?

Example - Relations properties

Example 2.8: Properties of Relations

Is the following relation reflexive, symmetric, antisymmetric, or transitive?

$$\mathcal{R} = \{(A, B) \in \mathcal{P}(\mathbb{Z}) \times \mathcal{P}(\mathbb{Z}) \mid A \subseteq B\}$$

Solution



Example - Relations properties

Example 2.9: Properties of Relations

Is the following relation reflexive, symmetric, antisymmetric, or transitive?

$$\mathcal{R} = \{(A, B) \in \mathcal{P}(\mathbb{Z}) \times \mathcal{P}(\mathbb{Z}) \mid A \subseteq B\}$$

Solution

Notice that A and B are both elements of $\mathcal{P}(\mathbb{Z})$. Therefore, they are each some subset of the integers, of any size. For example, $\{1, 2, 3, 4, 5\}$ and $\{1, 6, 23, 123, 1299, 1300\}$

- For any A we have $A \subseteq A$, so $\mathcal{R}$ is reflexive
- Consider $A = \{1\}$ and $B = \{1, 2\}$. $A \subseteq B$ but $B \not\subseteq A$. Therefore, $\mathcal{R}$ is not asymmetric
- We have $A \subseteq B \wedge B \subseteq A \rightarrow A = B$, so $\mathcal{R}$ is antisymmetric.
- For any sets A, B, C such that $A \subseteq B$ and $B \subseteq C$, it also holds that $A \subseteq C$. Therefore, $\mathcal{R}$ is transitive.



Equivalence Relations

- Equivalence relations are special kinds of relations with several simultaneous properties.
- Equivalence classes are essential in discrete mathematics and computer science.

Definition 2.6: Equivalence Relation

A binary relation $\mathcal{R}$ on a set A is an equivalence relation if it is reflexive, symmetric and transitive

- Recall that for a binary relation $\mathcal{R}$ we write $a\mathcal{R}b$ when $(a, b) \in \mathcal{R}$.
- When $\mathcal{R}$ is an equivalence relation, we can use the stronger notation $a \sim b$ to denote that a “is equivalent to” b under the equivalence relation $\mathcal{R}$.



Equivalence Relations

To prove a set is equivalent, a proof for Reflexivity, Symmetry and Transitivity is key. Therefore to be explicit, for a binary relation $\mathcal{R}$ on a set A we must prove

- ① Reflexive: $a \sim a \ \forall a \in A$.
- ② Symmetric: $a \sim b \rightarrow b \sim a$, and
- ③ Transitive: $a \sim b \wedge b \sim c \rightarrow a \sim c$.

Example 2.10: Postal codes, an equivalence relation

Consider the set of all people with a mailing address in JKUAT. Consider two people in this set to be related if they share a postal code. This relation is clearly an equivalence relation.

- ① **Reflexive.** Every person shares the same postal code as themselves.
- ② **Symmetric.** If two people x and y share a postal code, then y and x share a postal code.
- ③ **Transitive.** If a and b share a postal code, and b and c share a postal code, then a and c share a postal code.



Equivalence Classes

- An important consequence of equivalence relations are equivalence classes.
- Equivalence classes group together all elements which are equivalent under a certain equivalence relation.

Definition 2.7: Equivalence Class

Let $\sim$ denote an equivalence relation on a set A . Then, for an element $a \in A$, the equivalence class of a is the set $\{b \in A \mid b \sim a\}$

For an equivalence class $a \in A$ under some equivalence relation, equivalence classes are denoted in two possible ways, both using any representative of the class

$$[a] = \{b \in A \mid b \sim a\}$$

$$\bar{a} = \{b \in A \mid b \sim a\}$$



Equivalence Classes

Example 2.11: Equivalence classes on the integers

Let $\mathcal{R}$ be the equivalence relation $\{(a, b) \in \mathbb{Z}^2 \mid |a| = |b|\}$, result in the following equivalence classes

$$[1] = [-1] = \{-1, 1\}$$

$$[5] = [-5] = \{-5, 5\}$$

$$[49] = [-49] = \{-49, 49\}$$

In general, the equivalence classes under $\mathcal{R}$ are the sets $\{-i, i\}$ for all $i \in \mathbb{Z}^+$ and $[0]$

An equivalence relation on a set A creates a partition of X

Definition 2.8: Partition

A partition of a set A is a collection of disjoint subsets of A such that their union is all of A .



Example Partitions

Example 2.12: Example Partitions

- 1 The countries of the world are partitioned into continents.
- 2 A deck of cards is partitioned into four suits: hearts, diamonds, clubs, spades.
- 3 The integers are partitioned into odd integers and even integers (yes, 0 is even).
- 4 The integers can also be partitioned into three groups: positive, negative, and 0
- 5 The integers can also be partitioned into groups of pairs: $\{-i, i\}$ for any non-zero integer i , and a lone group $\{0\}$

Theorem 2.1

For an equivalence relation $\mathcal{R}$ on a set A , its set of equivalence classes form a partition of A



```
d = {}
for i in range(0,100) :
    if i % 10 in d:
        d[i % 10].append(i)
    else :
        d[i % 10] = [i]
#print the list
for (k,v) in d.items() :
    print(k, v);
```

```
0 [0, 10, 20, 30, 40, 50, 60, 70, 80, 90]
1 [1, 11, 21, 31, 41, 51, 61, 71, 81, 91]
2 [2, 12, 22, 32, 42, 52, 62, 72, 82, 92]
3 [3, 13, 23, 33, 43, 53, 63, 73, 83, 93]
4 [4, 14, 24, 34, 44, 54, 64, 74, 84, 94]
5 [5, 15, 25, 35, 45, 55, 65, 75, 85, 95]
6 [6, 16, 26, 36, 46, 56, 66, 76, 86, 96]
7 [7, 17, 27, 37, 47, 57, 67, 77, 87, 97]
8 [8, 18, 28, 38, 48, 58, 68, 78, 88, 98]
9 [9, 19, 29, 39, 49, 59, 69, 79, 89, 99]
```

Figure 11: “The equivalences classes under $\mathcal{R}$ are: $[0]$, $[1]$, $[2]$, $[3]$, $[4]$, $[5]$, $[6]$, $[7]$, $[8]$, $[9]$ ”



Partially Ordered Sets

Equivalence relations could be viewed as a weakened form of equality. The same can be said for inequality or ordering.

Proposition 2.1: Properties of $\leq$

Less than or equal to ($\leq$) is the typical order relation. Its properties are reflexivity, antisymmetry, and transitivity. Let's prove that.

Proof.

- 1 **Reflexive:** For any number $x \in \mathbb{R}$, $x \leq x \therefore$ it is reflexive
- 2 **Antisymmetric:** For any two numbers $x, y \in \mathbb{R}$ if $x \leq y$ and $y \leq x$, then surely $x = y \therefore$ it is antisymmetric
- 3 **Transitive:** For any three integers $x, y, z \in \mathbb{R}$ if $x \leq y$ and $y \leq z$ then $x \leq y \leq z$ and $x \leq z \therefore$ it is transitive



More generally, a binary relation defines a partial order if it is reflexive, antisymmetric, and transitive.



Partial Order

Definition 2.9: Partial Order

A binary relation $\mathcal{R}$ on a set A is a partial order if it is reflexive, antisymmetric, and transitive

- Recall $a \sim b$ notation denotes $a\mathcal{R}b$ for an equivalence relation $\mathcal{R}$
- Partial relation is denoted by $\preceq$, and use $a \preceq b$ to denote that the pair (a, b) belongs to the partial order, meaning that " a is less than b "



Exercises

- 1 Consider the set B of all Kenyan Bands(i.e. Musical Artists) and S of all students at JKUAT. How can we interpret the Cartesian Product $S \times B$. What is a possible binary relation that we could construct from S to B



Exercises

- ① Consider the set B of all Kenyan Bands(i.e. Musical Artists) and S of all students at JKUAT. How can we interpret the Cartesian Product $S \times B$. What is a possible binary relation that we could construct from S to B

Solution

$S \times B$ is all possible pairs (s, b) of a students s at JKUAT and Kenyan artist b . The possible relations from S to B Could be

- Student s enjoys the music from of band b
 - Student s has seen band b live in concert
 - Student s dislikes band b
 - Student s is a member of band b
- ② Consider the set S of different sports. Consider the set P of all people in Nairobi, Kenya

What is a possible binary relation from S to P ? What is a possible binary relation from P to S ? Can you think of a binary relation between these sets such that the ordering of coordinates is important? That is, $\mathcal{R} \subseteq S \times P$ has a different meaning if we constructed it as $\mathcal{R} \subseteq P \times S$



Exercises

Solution



Exercises

Solution

S to P : relate $s \in S$ to $p \in P$ where s is a sport that p likes to participate in. P to S relate $p \in P$ to $s \in S$ where p is on a team for sport s . No such relation is immediately obvious. Generally speaking, binary relations from one set to a different set are symmetric. For example " p participates in sport s " and " s has p as a participant" are the same; (p, s) and (s, p) can be interpreted as the same thing.

- 4 Let $A = \{0, 2, 5, 6, 7\}$ and let $B = \{1, 2, 3, 4, 6\}$

List the pairs contained in the following relations.

- $\mathcal{R}_1 = \{(a, b) \in A \times B \mid a = b\}$
- $\mathcal{R}_2 = \{(a, b) \in A \times B \mid a < b\}$
- $\mathcal{R}_3 = \{(a, b) \in A \times B \mid a \geq b\}$
- $\mathcal{R}_4 = \{(a, b) \in A \times B \mid a - b = 2\}$



Exercises

Solution



Exercises

Solution

$$\mathcal{R}_1 = \{(2, 2), (6, 6)\}$$

$$\mathcal{R}_2 = \{(0, 1), (0, 2), (0, 3), (0, 4), (0, 6), (2, 3), (2, 4), (2, 6), (5, 6)\}$$

$$\mathcal{R}_3 = \{(7, 1), (7, 2), (7, 3), (7, 4), (7, 6), (6, 1), (6, 2), (6, 3), (6, 4), \\ (6, 6), (5, 1), (5, 2), (5, 3), (5, 4), (2, 2), (2, 1)\}$$

$$\mathcal{R}_4 = \{(5, 3), (6, 4)\}$$

- ⑥ Let $A = \{x \in \mathbb{Z}^- | x > -5\}$ and Let $B = \{x \in \mathbb{N} | x < 8\}$

List the pairs contained in the following relations

- $\mathcal{R}_1 = \{(a, b) \in A \times B \mid |a| = |b|\}$
- $\mathcal{R}_2 = \{(a, b) \in A \times B \mid -a > b\}$
- $\mathcal{R}_3 = \{(a, b) \in A \times B \mid 3a + 2b = 0\}$
- $\mathcal{R}_4 = \{(a, b) \in A \times B \mid b < a\}$



Solution



Solution

$$\mathcal{R}_1 = \{(-1, 1), (-2, 2), (-3, 3), (-4, 4)\}$$

$$\mathcal{R}_2 = \{(-4, 0), (-4, 1), (-4, 2), (-4, 3), (-3, 0), (-3, 1), \\ (-3, 2), (-2, 0), (-2, 1), (-1, 0)\}$$

$$\mathcal{R}_3 = \{(-2, 3), (-4, 6)\}$$

$$\mathcal{R}_4 = \emptyset$$

- 8 Let $A = \{0, 2, 5, 6, 7\}$ and Let $B = \{1, 2, 3, 4, 6\}$
Draw a table representing the relation:

$$\mathcal{R}_2 = \{(a, b) \in A \times B \mid a < b\}$$



Exercises

- 9 Consider binary relations on the set $A = \{a, b, c, d\}$ Are the following relations reflexive? Symmetric? Antisymmetric? Transitive?
- i $\mathcal{R}_1 = \{(b, b), (b, c), (b, d), (c, b), (c, c), (c, d)\}$
 - ii $\mathcal{R}_2 = \{(a, a), (a, b), (b, a), (b, b), (c, c), (d, d)\}$
 - iii $\mathcal{R}_3 = \{(d, b), (b, d)\}$
 - iv $\mathcal{R}_4 = \{(a, b), (b, c), (c, d)\}$
 - v $\mathcal{R}_5 = \{(a, a), (b, b), (c, c), (d, d)\}$
 - vi $\mathcal{R}_6 = \{(a, c), (a, d), (b, c), (b, d), (c, a), (c, d)\}$

Solution



Exercises

- 9 Consider binary relations on the set $A = \{a, b, c, d\}$ Are the following relations reflexive? Symmetric? Antisymmetric? Transitive?
- i $\mathcal{R}_1 = \{(b, b), (b, c), (b, d), (c, b), (c, c), (c, d)\}$
 - ii $\mathcal{R}_2 = \{(a, a), (a, b), (b, a), (b, b), (c, c), (d, d)\}$
 - iii $\mathcal{R}_3 = \{(d, b), (b, d)\}$
 - iv $\mathcal{R}_4 = \{(a, b), (b, c), (c, d)\}$
 - v $\mathcal{R}_5 = \{(a, a), (b, b), (c, c), (d, d)\}$
 - vi $\mathcal{R}_6 = \{(a, c), (a, d), (b, c), (b, d), (c, a), (c, d)\}$

Solution

- i Transitive.
- ii Reflexive, symmetric, and transitive.
- iii Symmetric.
- iv Antisymmetric (vacuously so).
- v Reflexive, symmetric, transitive, and antisymmetric.
- vi Nothing.